

HAT-P-36b

R-0887

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240224 · created 2026-07-29T01:18:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460364.866 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1573 +/- 0.0052
Transit depth (Rp/Rs)^2	2.47 +/- 0.16 [%]
Orbital Inclination (inc)	85.08 +/- 2.07
Airmass coefficient 1 (a1)	1.366 +/- 0.024
Airmass coefficient 2 (a2)	-0.295 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	None
Optimal Aperture	3.55
Optimal Annulus	11.05
Transit Duration (day)	0.0942 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1573 ± 0.0052 vs 0.126 (deviation 5.89σ)
Duration fit vs published	0.0942 d vs 0.0925 d (deviation 0.33σ)
Mid-time O–C	-2.9 min
Depth SNR	29.6
Residual scatter	1.73 %

Light curve

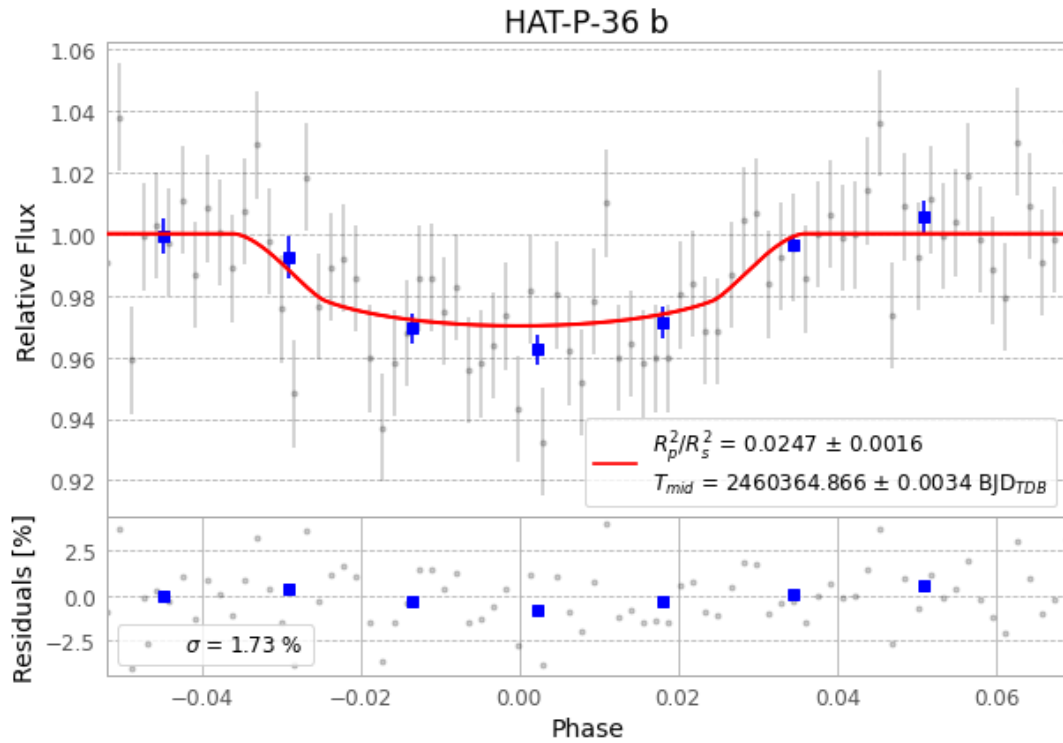


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-02-24.png (SHA-256 623dc10141f9671b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f2c541807966391c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), HATP-36240224070015.FITS → HATP-36240224105116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240224.log (d5897938a394...), FinalLightCurve_HAT-P-36 b_2024-02-24.png (623dc10141f9...), FinalLightCurve_HAT-P-36 b_2024-02-24.csv (28349995c39b...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:18Z.